

MDM2 – Case Study: Assistive Robotics for Improvement of Life Quality One-Page Proposal

Team	Group 8
Members	Prajwal Dikonda, Nishant Shetty, Papiha Dhavan
Project Title	EEG-Based Brain-Computer Interface for Assistive Robotic Arm Control
GitHub Repository URL	https://github.com/dikondaprajwal/eeg-bci-controlled-robotic-arm.git
Contact Email	dikonda.prajwal@outlook.com
Industrial Application (target domain/use-case)	Healthcare & Assistive Robotics (Neuro-controlled systems for individuals with motor impairments)
Keywords (3–6)	Brain–Computer Interface (BCI), EEG, Assistive Robotics, Human–Robot Interaction, Robotic Arm Control
Submission Date (YYYY-MM-DD)	2026-04-10
Gantt Chart	Project Gantt Chart The Gantt chart illustrates the project schedule with the following phases and durations: P1: Proposal: April 10 to April 24, 2026. P2: Literature + Dataset: April 24 to May 15, 2026. P3: Processing + ML: May 15 to June 12, 2026. P4: Simulation + Integration: June 12 to July 3, 2026. P5: Testing + Report: July 3 to July 10, 2026.

1) Problem Statement & Measurable Outcomes (3–4 sentences)	Millions of individuals with paralysis or motor impairments are unable to perform basic daily activities despite having intact cognitive function. Approximately 15 million strokes occur annually, with around 5 million resulting in permanent disability. This project aims to develop an EEG-based brain–computer interface that enables users to control a robotic arm using brain signals. KPIs: - Classification accuracy $\geq 75\%$ - Response latency ≤ 500 ms - Command success rate $\geq 80\%$
---	---

	- Ability to perform at least 4–5 basic arm movements
2) Motivation & Assistive Relevance (2–3 sentences)	Assistive technologies are essential for improving independence and quality of life for individuals with severe motor impairments. With an aging population and increasing cases of neurological disorders, there is a growing need for intuitive, non-physical control systems.
3) Related Work Snapshot (2–3 key references)	- EEG-based brain–computer interfaces for robotic control - Assistive robotic arms for rehabilitation and daily living tasks - Signal processing and machine learning techniques for neural signal classification Gap: Existing systems are complex, expensive, and difficult to deploy. This project focuses on a simplified and scalable implementation.
4) Method & Feasibility (≤6 sentences)	The system will use EEG data (dataset or consumer device). Signal preprocessing techniques such as band-pass filtering will remove noise. Features will be extracted and classified using simple ML models (SVM or Logistic Regression). The output will control a robotic arm. Expected Outputs: - EEG signal processing pipeline - Intention classification model - Command mapping system - Robotic arm simulation or prototype
5) Milestones & Timeline (short table/list)	- 10 April: Proposal submission - 24 April: Literature review and dataset selection - 15 May: Signal processing and classification - 12 June: System integration and simulation - 03 July: Final report and presentation
6) Risks & Ethics (1–2 sentences)	Potential risks include noisy EEG signals, low classification accuracy, and variability between users. Ethical considerations include safe system behavior and responsible use of assistive technology.

Phase 1 rubric (15%): Team & GitHub (2%), On-time (2%), Topic & Proposal (5%) — Industrial Application, Problem+Outcomes, Feasibility+Timeline; Presentation (6%).